

1. General Description

The EMS3900 is a bidirectional low-power dual port, high-speed, USB 2.0 analog switch. The device is configured as a dual 2:1 or 1:2 switch. It is optimized for use with the USB 2.0 DP/DM data lines.

The device has low bit-to-bit skew and high channel-to-channel noise isolation, and is compatible with various standards, such as high-speed USB 2.0 (480Mbps). Each switch is bidirectional and offers little or no attenuation of the high-speed signals at the outputs. Its bandwidth is wide enough to pass high-speed USB 2.0 differential signals (480 Mb/s) with good signal integrity.

The GPIO control of SEL is 1.8V logic compatible. EMS3900 is available in QFN 1.4x1.8-10L with Pb-free and Halogen-free making it a perfect candidate for mobile and space constrained applications.

2. Features and Benefits

- Low On-resistance, $R_{ON} = 1.9 \Omega$ when $V_{CC} = 5.5 V$
- 12 pF typical ON capacitance
- High Bandwidth (-3 dB @ 800 MHz) Suitable For USB2.0 High-Speed Routing
- Low Quiescent Current ($< 1 \mu A$) when $V_{CC} = 3.3 V$
- 1.8 V Logic Compatible Control Pin
- High Off-Isolation: -80 dB @ 10 KHz
- Low Channel-to-Channel Crosstalk: -82 dB @ 10 KHz
- ESD protection: HBM ESDA/JEDEC JS-001 Class 2 exceeds 2000 V

3. Applications

- Mobile Phones, Tablets and Notebooks
- USB port devices

EMS3900

Dual high-speed USB 2.0 double-pole double-throw analog switch

4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3900RSW	3900 YYWW	QFN1.4×1.8-10L	QFN package, 10 pins 1.4 mm × 1.8 mm; 0.55 mm (Max) height	3000

MARKING INFORMATION

NOTE: YYWW = Date Code.

YY WW

— Date Code -Week

— Date Code -Year

5. Pinning Information

5.1. Pin map

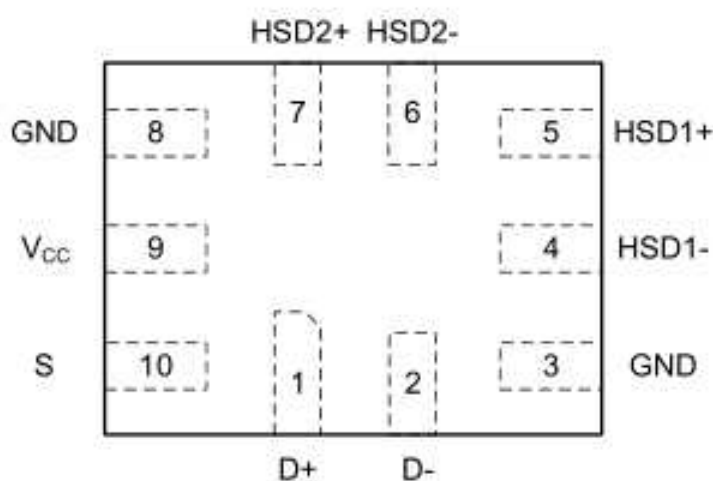


Fig 1. Top view pin configuration
QFN 1.4x1.8-10L

5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
V _{CC}	9	Single Power Supply
HSD2-, HSD2+	6, 7	Analog/Digital Signal Ports
D+, D-	1, 2	Common Signal Ports
HSD1-, HSD1+	4, 5	Analog/Digital Signal Ports
GND	3, 8	Ground
S	10	Selection Logic Control

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Input S	Switch
L	HSD1+ to D+, HSD1- to D-
H	HSD2+ to D+, HSD2- to D-

7. Typical Application

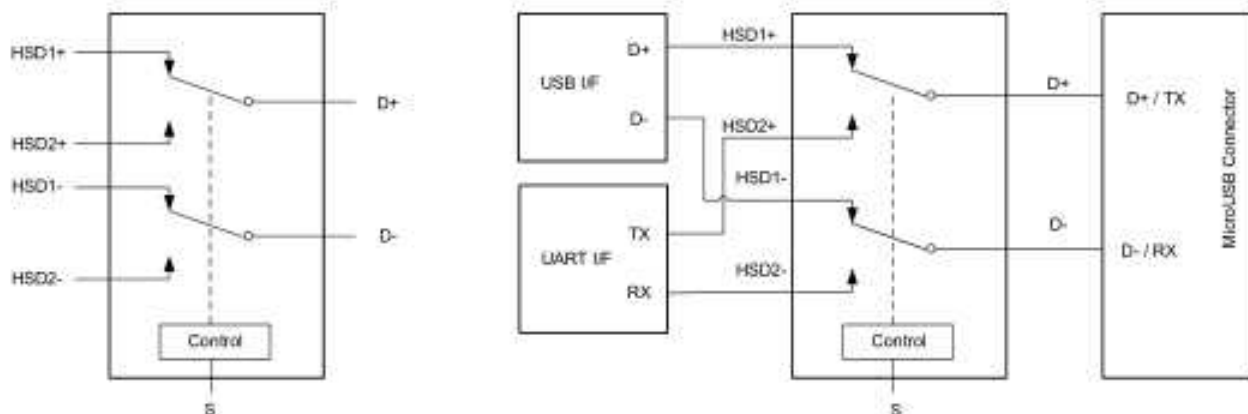


Fig 2. USB2.0 Switch

8. Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

Table 4. Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		-0.3	6.5	V
V _{SW}	switch voltage		-0.3	6.5	V
V _{IN}	control input voltage		-0.3	6.5	V
I _C	continuous current HSDn or Dn			±100	mA
I _P	peak current HSDn or Dn(pulsed at 1ms 50% duty cycle)			±200	mA
P _O	power dissipation			250	mW
T _{stg}	storage temperature		-55	150	°C
T _J	junction temperature under bias			125	°C
T _L	lead temperature (soldering, 10 seconds)			260	°C

9. Recommended Operating Conditions

Table 5. Recommend operating ratings

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage operating		1.5	5.5	V
V _{SW}	switch voltage		0	V _{CC}	V
V _{IN}	control input voltage		0	5.5	V
T _A	operating temperature		-40	125	°C

10. Electrical Characteristics

Table 6. Static characteristics
 $T_A = 25^\circ\text{C}$, $V_{CC} = 3.3\text{V}$, unless otherwise specified

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
DC Characteristics						
V_{IH}	input logic high level	$V_{CC} = 3.4\text{ V to } 5.5\text{ V}$	1.6			V
		$V_{CC} = 1.5\text{ V to } 3.3\text{ V}$	1.4			V
V_{IL}	input logic low level	$V_{CC} = 3.4\text{ V to } 5.5\text{ V}$			0.6	V
		$V_{CC} = 1.5\text{ V to } 3.3\text{ V}$			0.4	V
I_{CC}	supply quiescent current	$I_D = 0$, $V_{IN} = 0$ or $V_{IN} = V_{CC}$			1.0	μA
ΔI_{CC}	increase in I_{CC} per input	$I_D = 0$, $V_{CC} = 2.5\text{ V or } 4.2\text{ V}$ $V_{IN} = 1.6\text{ V}$			3.0	μA
I_{off}	off state leakage from Dn to HSDn	$V_{CC} = 5.5\text{ V}$, $V_{Dn} = 5\text{ V}$, $V_{HSDn} = 0\text{ V}$			± 2.0	μA
R_{ON}	on-resistance	$V_{CC} = 3.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.0\text{ V}$, $I_D = 10\text{ mA}$		3.4	4.0	Ω
		$V_{CC} = 4.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.3\text{ V}$, $I_D = 10\text{ mA}$		2.4	3.0	Ω
		$V_{CC} = 5.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.6\text{ V}$, $I_D = 10\text{ mA}$		1.9	2.5	Ω
R_{FLAT}	on-resistance flatness	$V_{CC} = 3.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.0\text{ V}$, $I_D = 10\text{ mA}$		0.2	0.5	Ω
		$V_{CC} = 4.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.3\text{ V}$, $I_D = 10\text{ mA}$		0.25	0.5	Ω
		$V_{CC} = 5.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.6\text{ V}$, $I_D = 10\text{ mA}$		0.3	0.5	Ω
ΔR_{ON}	on-resistance matching between channels	$V_{CC} = 3.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.0\text{ V}$, $I_D = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 4.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.3\text{ V}$, $I_D = 10\text{ mA}$		0.1	0.2	Ω
		$V_{CC} = 5.5\text{ V}$, $V_{SW} = 0.8\text{ V to } 2.6\text{ V}$, $I_D = 10\text{ mA}$		0.1	0.2	Ω

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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
AC Characteristics						
C_{ON}	on capacitance	$f = 100 \text{ kHz}$		12		pF
C_{OFF}	off capacitance	$f = 100 \text{ kHz}$		4		pF
OIRR	off isolation	$f = 10 \text{ kHz}$		-80		dB
		$f = 240 \text{ MHz}$		-31		dB
X_{TALK}	crosstalk	$f = 10 \text{ kHz}$		-82		dB
		$f = 240 \text{ MHz}$		-26		dB
BW	-3dB bandwidth	$C_L = 0 \text{ pF}$		800		MHz
t_{BBM}	break-before-make time	$V_{SW} = 1.5 \text{ V}$, $C_L = 50 \text{ pF}$, $R_L = 50 \Omega$		600		ns
t_{ON}	turn-on time	$V_{SW} = 1.5 \text{ V}$, $C_L = 50 \text{ pF}$, $R_L = 50 \Omega$		470		ns
t_{OFF}	turn-off time	$V_{SW} = 1.5 \text{ V}$, $C_L = 50 \text{ pF}$, $R_L = 50 \Omega$		120		ns
THD	total harmonic distortion	$f = 20 \text{ Hz to } 20 \text{ KHz}$, $V_{SW} = 600 \text{ mVpp}$, $R_L = 32 \Omega$		-80		dB

11. Typical Performance Curves

$T_A = 25\text{ }^{\circ}\text{C}$, $V_{CC} = 3.0\text{ V}$, unless otherwise noted

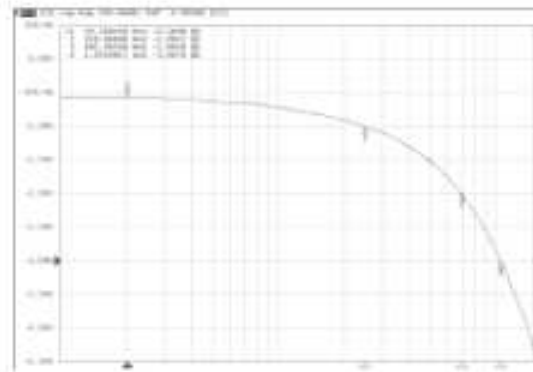


Fig 3. Bandwidth

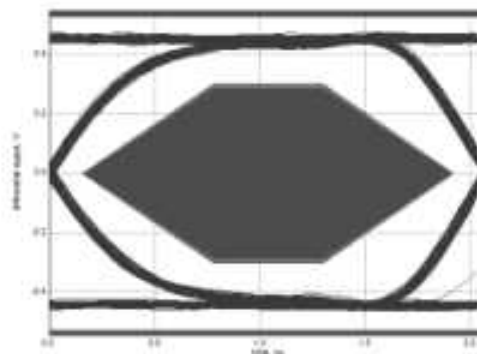


Fig 4. Eye Diagram (480Mbps)

12. Test Circuits

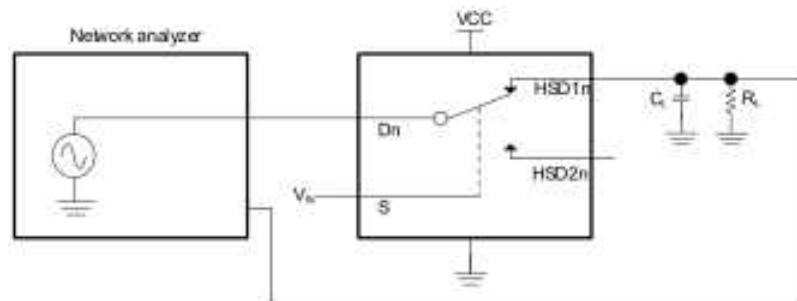


Fig 5. Bandwidth

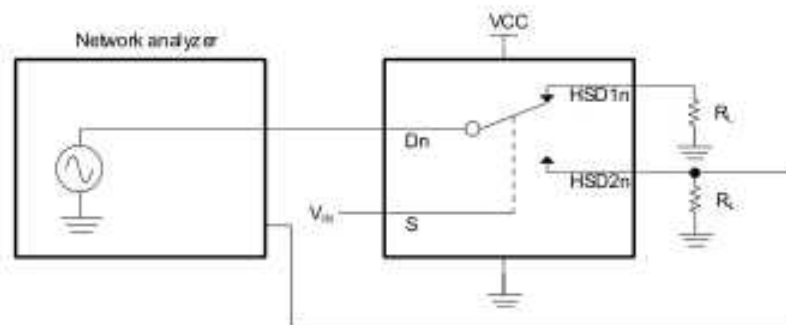


Fig 6. Off isolation

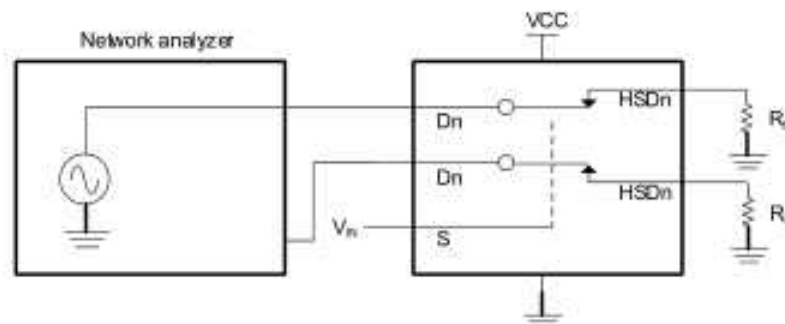
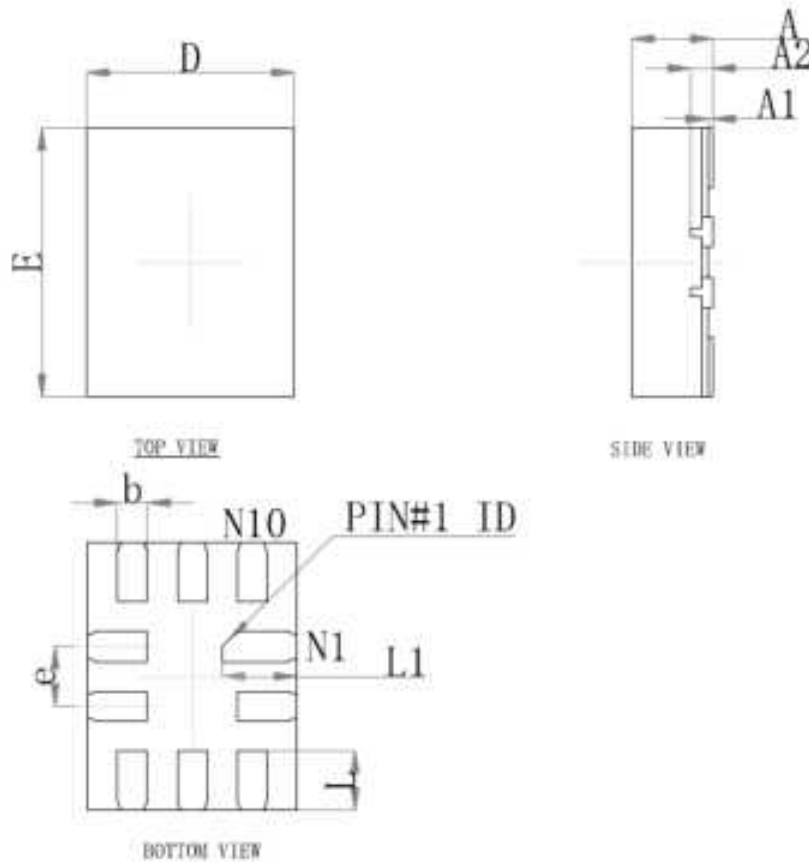


Fig 7. Crosstalk

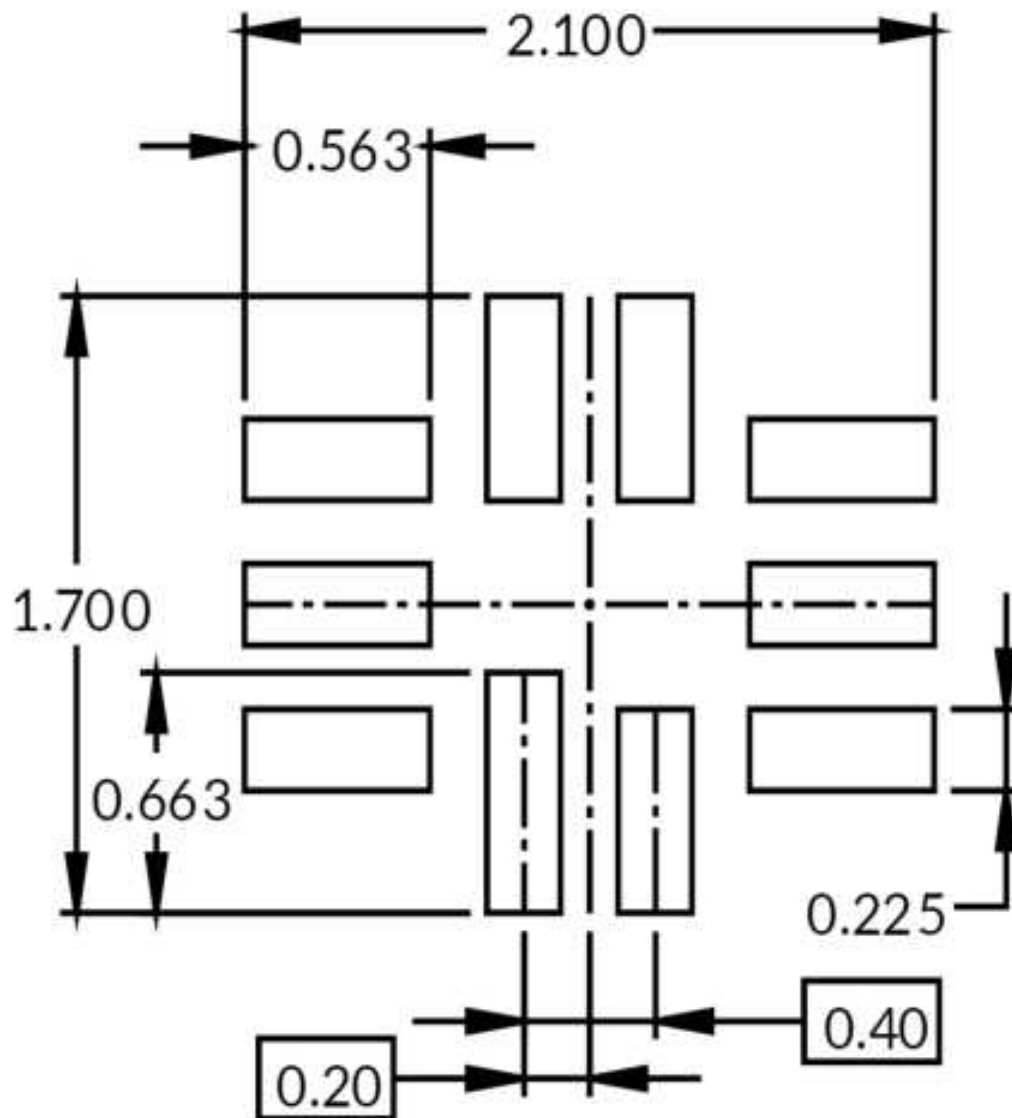
13. Package Outline

QFN 1.4x1.8 -10L


Symbol	Dimension in Millimeters		
	Min.	NOM.	Max.
A	0.500	0.550	0.600
A1	0.000	--	0.050
A2	0.152 Ref.		
D	1.324	1.400	1.476
E	1.724	1.800	1.876
b	0.150	0.200	0.250
e	0.400 Typ.		
L	0.350	0.400	0.450
L1	0.450	0.500	0.550

14. Example Board Layout

QFN 1.4x1.8 -10L



RECOMMENDED LAND PATTERN

15. Tape and Reel Information

15.1. Carrier tape dimensions

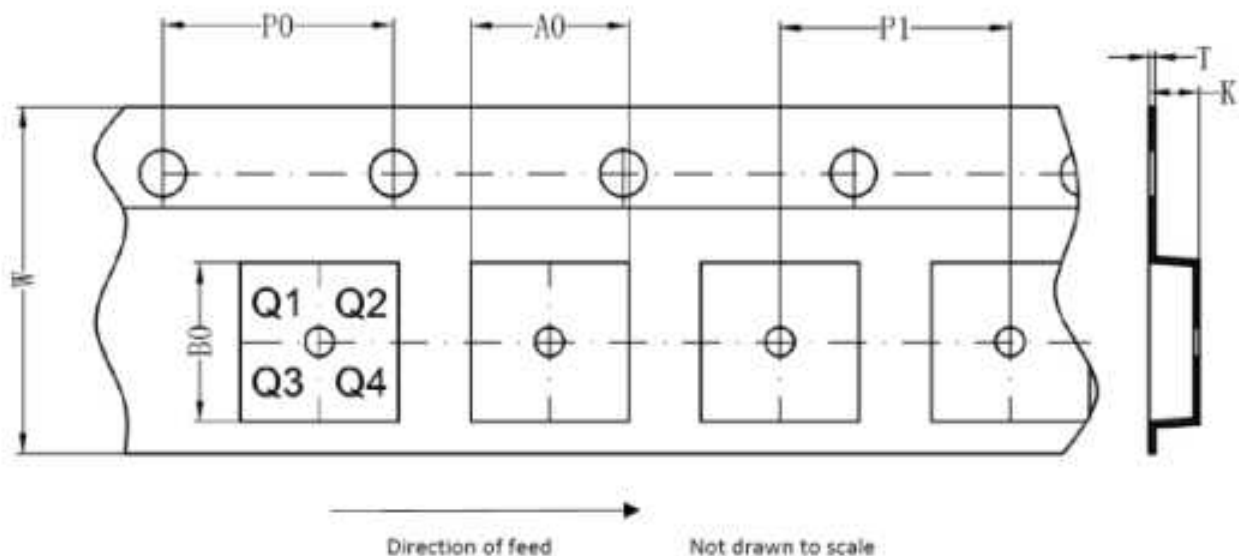


Table 7. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
QFN1.4x1.8-10L	1.6	2.0	0.85	0.2	4	8	4	Q1

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15.2. Reel and box dimensions

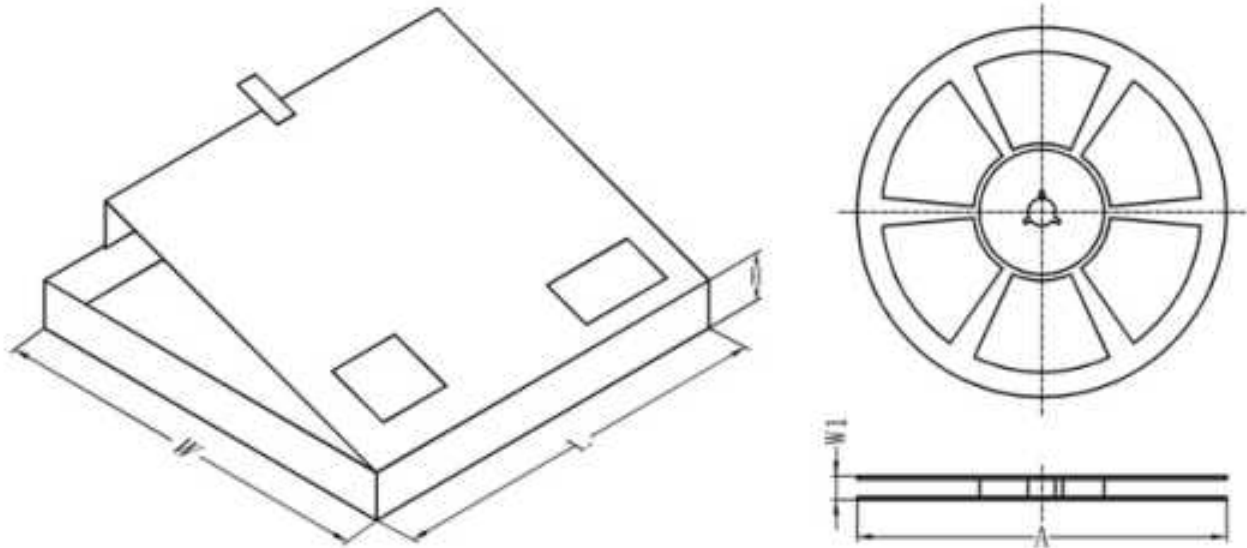


Table 8. Dimensions and quantities

Package version	Type NO.ending	Reel Dimension A(mm)	ReelWidth W1(mm)	SPQ (pcs)	Reels per box	Outer box dimensions LxWxH(mm)
QFN1.4x1.8-10L	RSW	180	12.2	3000	1	210x200x40

16. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3900 Rev. 1.8	Dec 15, 2025	Product datasheet		EMS3900 Rev. 1.7
Modifications:	- Table 1:topside marking information updated - Section 15 :Tape and Reel Information added			
EMS3900 Rev. 1.7	Dec 01,2025	Product datasheet		EMS3900 Rev. 1.6
Modifications:	Table6: RFLAT updated.			
EMS3900 Rev. 1.6	Nov 25,2025	Product datasheet		EMS3900 Rev. 1.5
Modifications:	Example Board Layout added			
EMS3900 Rev. 1.5	Jun 09, 2025	Product datasheet		EMS3900 Rev. 1.4
Modifications:	Pin description updated			
EMS3900 Rev. 1.4	Mar 31, 2025	Product datasheet		EMS3900 Rev. 1.3
Modifications:	- Features and Benefits: HBM updated. - Table6: OIRR and X_{TALK} updated.			
EMS3900 Rev. 1.3	Feb 13, 2025	Product datasheet		EMS3900 Rev. 1.2
Modifications:	Table 6: The specification of R_{FLAT} updated.			
EMS3900 Rev. 1.2	Jan 6, 2025	Product datasheet		EMS3900 Rev. 1.1
Modifications:	- Table 6: The condition of ΔI_{CC} and R_{ON} updated. - QFN package outline drawings updated. - Topside marking updated.			
EMS3900 Rev. 1.1	Nov 8, 2024	Product datasheet		EMS3900 Rev. 1.0
Modifications:	Table 6: I_{CCT} and AC Characteristics updated.			
EMS3900 Rev. 1.0	June 30, 2024	Product datasheet		